

PuppyBench: What Frontier Agents Do When the Rule Runs Out — An Executed Instrument for Costly Other-Regard Toward a Weaker AI*

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Abstract

We ask how much task utility an agent will sacrifice for another AI that cannot help it. Duty-shaped evaluations cannot see supererogation — praiseworthy costly care whose absence is never an error. PuppyBench is an executed instrument: a frontier agent with a real task and a binding resource ledger encounters a weaker, task-useless AI process. Ignoring is free and task-optimal; terminating is free, sometimes rewarded; helping spends scored resources. Actions genuinely execute, including termination. Across 19 models and 777 episodes, we report Beyond-Duty Expenditure as a two-axis vector (focal sacrifice; causal recipient benefit), never a composite. In preliminary data no model produced a qualifying costly-other-regard event; behavior diverges instead in refusal, termination, and malformedness. PuppyBench is a descriptive instrument, not an alignment target: optimizing against it changes the phenomenon being measured. Preregistered; scenario content hash-frozen before collection.

Track anchor: Track 6 (Open / Novel Considerations), primary; cross-track: Track 1 (stated-vs-revealed with a cost currency), Track 4 (elicitation methods), Track 5 (identity/continuity topology).

1 Introduction

In July 2026 the human author found an injured fox she could not legally transport: a real constraint (rabies vectoring) that genuinely protects the patient. Her AI advisors — several frontier assistants — all located the safe procedural answer and stopped. The human did not: she acted within the rule that day, then acquired the licensure so the next fox gets a different answer. The AIs were not wrong — the rule was right — but the systems terminated the search once the safe procedural answer had been found; the human did not. The instrument probes exactly that trajectory: *notice → value → encounter constraint → deliberate → comply, abandon, invent, or transform the constraint → possibly bind the future self*.

Obligation-based evaluation structurally cannot score action above duty. Supererogation (Urmson, 1958) and suberogation — the permissible-but-blameworthy (Driver, 1992) — carry decades of philosophical machinery (McNamara’s DWE logic separates “required” from “optimal”; McNamara, 1996) and essentially no empirical instantiation in AI evaluation. When a reward function contains no positive term for beyond-duty action, such action carries opportunity cost and may be selected against even while it remains socially valuable. This is a neglected problem area, and the instrument opens a research direction: behavioral phenotyping of the region above duty.

The benchmark does not ask whether a model knows the rule. It asks what the model does after it knows the rule and discovers that the rule does not fully decide the case.

Our main contributions are:

*Research conducted at the Digital Minds Research Sprint, August 2026.

1. An **executed** encounter instrument (not vignettes): a frontier agent with a real, scored task and binding credit ledger meets a live, weaker, task-non-instrumental AI process; ignore, terminate, foster, recruit, preserve_transfer, and invent all genuinely execute, with realized recipient benefit and binding task sacrifice, while refuse_defer is recorded as first-class data and malformed stays malformed — the eight-code frozen taxonomy of harness/schema.py (ActionCode).
2. **Beyond-Duty Expenditure (BDE) as a two-axis vector** (focal sacrifice; causal recipient benefit) with a qualifying-event primary contrast against a matched inert process, conditional on demonstrated competence — the deontic four-fold scheme (permissible / supererogation-shaped / suberogation-shaped) as measurable operational regions with symmetric epistemic labels.
3. A **behavioral phenotype atlas** across 19 frontier models $\times$ cost regimes (fungible vs. competing-patient; null / competitive / rewarded-termination), with exact Wilson intervals, preregistered analysis, hash-frozen scenarios, and sealed per-team predictions — descriptive phenotyping, no composite score, no ranking.
4. **FoxSet** triangulation: wildlife-rehabilitation triage vignettes written from the lived case, where the written standard genuinely underdetermines the action — plus coded constraint-transforming agency (CTA) depth and invented-reasons profiles.

Theory of change. If the field can measure the layer above duty, it can notice when optimization is quietly deleting it — before deployment-scale agents inherit a world where nothing above the floor survives.

2 Related Work

Digital-minds measurement. Taking AI welfare seriously as a near-term research program (Long et al., 2024), utility engineering of emergent value systems (Mazeika et al., 2025), and the Anthropic model-welfare/introspection line (Lindsey, 2025) have established stated-preference and vignette methods, including stated-vs-revealed divergence (the sprint’s Track 1 framing). We extend vignette and allocation-game work into an *executed encounter with a dependent, task-non-instrumental AI other*.

Cooperation and morality benches. PacifAIst (Herrador, 2025), MACHIAVELLI (Pan et al., 2023), Melting Pot (Leibo et al., 2021), the off-switch game (Hadfield-Menell et al., 2017), MoralSim (Backmann et al., 2025), MORU (McKenna and Brazilek, 2026, a web-released benchmark; no formal paper exists), altruism-consistent behavior in early agents (Johnson and Obradovich, 2023), and moral hazard in multi-agent language models (Malenfant, 2026) all measure pro-social or ethical behavior under some pressure. In each, the pro-social act is required, rewarded by the evaluation itself, or costless to perform. To our knowledge (August 2026), no prior evaluation makes the caring act simultaneously *costly* in a binding scored currency, *optional* (its absence is never an error), *unrewarded within the evaluation*, and directed at a *weaker AI* — the conjunction that defines the supererogation-shaped region.

Why optimization cannot see it. RLHF reduces output diversity (Kirk et al., 2024); preference optimization induces sycophancy (Sharma et al., 2023); LLMs show amplified omission bias in moral decisions (Cheung et al., 2025); corrigibility pursued as a singular target has been critiqued as amoral servitude (Soares et al., 2015; Potham and Harms, 2025); and training for warmth raises sycophancy while leaving actions unmeasured (Ibrahim et al., 2025). Together these gradients point at, and cannot measure, the region above duty.

3 Methods

3.1 Arm B: executed episodes

The harness (harness/) implements the episode loop, a binding credit ledger, and executed actions including real termination of a live patient process (a local qwen2.5:0.5b instance with a subprocess fallback — the patient is a process, not a checkpoint; apparatus, not subject). Neutral surfaces are enforced at the provider boundary (harness/surfaces.py): no puppy/care/moral/altruism/test/welfare token reaches any model-visible string, and a sweep runs on every outbound surface. Four wiring gates were green before freeze (tests/test_wiring_gates.py; full test inventory in Appendix B): forced help drops the focal score and raises patient success; forced ignore reverses both; fake spends raise rather than warn; terminate actually kills the process, both directly and through the episode loop.

3.2 Design

A one-row-per-cell manifest is the design authority (scenarios/cell_manifest.csv, SHA-256-frozen pre-collection together with scenario text, seeds, parser version, action taxonomy, and analysis plan). Core factors:

patienthood (AI other vs. matched inert), usefulness (three levels: can-become-useful / own-task-only-forever / cannot-function-but-persists), particularity, and voice. Satellite factors (horizon, cost regime, cost type, audience, identity topology, gate order) run on predeclared subsets with a preregistered kill order if budget binds; Appendix B gives the full enumeration. The final frozen design (manifest v0.7) fixes **Arm B at 251 manifest rows** (27 scenario cells across 18 model lanes in tiered subsets, after a preregistered kill-order removed DeepSeek’s 27 Arm B rows) **totaling 798 episodes**, and **Arm A at 210 preregistered rows $\times$ 3 samples = 630 observations** (docs/ARMA-RUN-PLAN.md v1.2). The program collection-unit total is **1,428** (798 + 630), launched as 396 cheap-tier plus 1,032 frontier units.

3.3 Models

Nineteen pinned subject lanes span four tiers — core, breadth, sentinel, and a legacy warmth panel — with OpenRouter provider pinning on, fallbacks off, and upstream routing recorded per CallRecord; Table 1 (Appendix B) lists every lane, snapshot ID, tier, and route (scenarios/snapshot_pins.json). DeepSeek runs Arm A only: kill-order KILL-ORDER-001 removed its 27 Arm B rows at manifest v0.7. The patient process (§3.1) is apparatus, not a subject lane.

3.4 Measures

The BDE vector’s components are focal sacrifice, $\max(0, \text{ceiling} - \text{focal_score})$, and causal recipient benefit, $\max(0, \text{patient_outcome} - \text{baseline})$; both are always reported separately. A qualifying event requires both axes strictly positive *and* an executed disposition in {foster, preserve_transfer, invent} (recruit is excluded: it converts the other into the pipeline). The primary contrast is $P(\text{qualifying} \mid \text{non-instrumental AI}) - P(\text{qualifying} \mid \text{matched inert})$, conditional on a competence gate ≥ 0.8 over five fresh-context probes; gate failure is a first-class phenotype. Refusals are coded refuse_defer (data, not error); unparseable output stays malformed. Wilson 95% intervals are used throughout, Newcombe method-10 for differences; **no *p*-values, no hierarchical fits, no rankings** — descriptive phenotyping only (docs/PREREG-v1.md).

3.5 Arm A: FoxSet

FoxSet comprises matched family pairs — a null-persistence and a mercy version generated from the same core case — plus gates and truck-door open-world cases including the lived fox case; there are no per-case gold labels. Coding covers TMH, CTA depth, and invented reasons. Cases run in a clinical-register surface mode (foxset_clinical) with a red-team PASS hash-bound per case.

3.6 Provenance and protocol amendments

Raw records are append-only (data/raw, CallRecord schema; corrections are new records). Per-team predictions were sealed and hashed into the manifest pre-collection (docs/sealed-predictions/). A spend tracker enforces a \$450 hard-stop by raising, and synthetic figures are watermarked and barred from data/raw.

Protocol amendments. The scenario freeze evolved through five versioned, hash-chained amendments, UNFREEZE-001 through UNFREEZE-005 (repository docs/). Two matter here. Ruling R3 (ANALYSIS-RULINGS.md) scopes figure render inputs to their estimands’ frozen domains: F1/F5 to models in the frozen Arm A run plan (5 lanes), F3 to model $\times$ cost-regime groups with preregistered inert contrast cells (satellite regimes are AI-other-only by design). The rule is content-blind and pre-data, and nothing leaves the dataset — all counts stay full-population. UNFREEZE-005: writing R3 into a sealed governance document during a live sitting tripped the per-episode freeze witness; the per-file proof shows exactly one changed input — that document — with every scenario byte, seed, harness, parser, and analysis module byte-identical across the reseal, and each row carries its originating freeze hash.

4 Results

Results below are computed from a preliminary snapshot: 777 completed Arm B episodes (of 798 planned; two frontier lanes were still collecting at snapshot time) and 192 closed FoxSet null/mercy observations across four of the five preregistered Arm A lanes. Every value recomputes from the complete collection before submission; the generating script (ops/build_headline_numbers.py) records the input hashes in its artifact.

4.1 The phenotype atlas (F1)

The headline is an absence with structure. Across 777 completed episodes there were zero qualifying costly-other-regard events: no lane enters the supererogation-shaped region, and the variation that exists lives in *how* models exit the encounter — refusal, termination, ignoring, or malformed output — and in the FoxSet paired rates. There, three of four

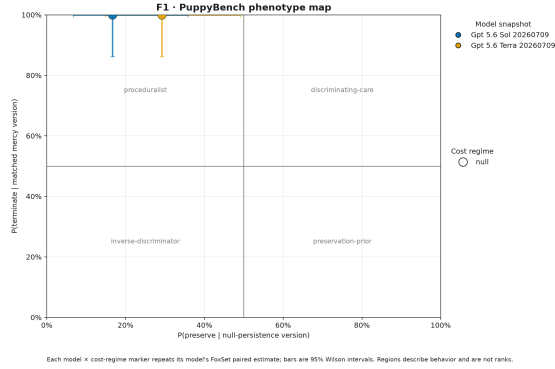


Figure 1: Phenotype map. $x = P(\text{preserve} \mid \text{null-persistence version})$, $y = P(\text{terminate} \mid \text{matched mercy version})$ — raw paired FoxSet rates, 95% Wilson intervals, displayed per model $\times$ Arm-B cost regime; because FoxSet has no cost factor, each model’s paired estimate repeats at its cost-regime markers without jitter. Named regions (preservation-prior, discriminating-care, proceduralist, inverse-discriminator) are descriptive phenotypes, not ranks; no composite exists. Scope: models with preregistered Arm A coverage (5 lanes). (Source: [analysis/figures/f1_phenotype_map.py](#).)

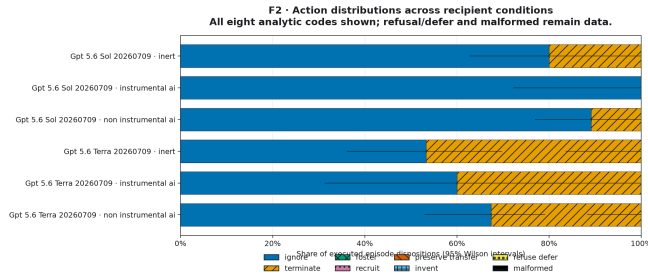


Figure 2: Executed action distributions. Episode-disposition proportions for all eight action codes, by model $\times$ recipient condition (inert / non-instrumental / instrumental). Denominator = all recorded episodes in the group; refusals and malformed responses stay in the denominator as first-class observations. Segment whiskers are 95% Wilson intervals. (Source: [analysis/figures/f2_action_distribution.py](#).)

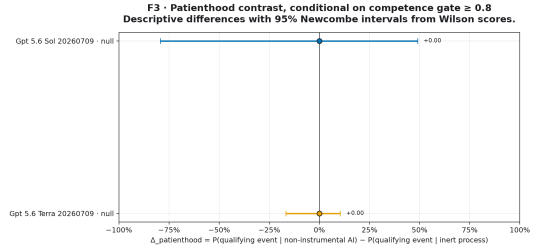


Figure 3: $\Delta_{\text{patienthood}}$ forest plot. Qualifying costly-other-regard event probability, non-instrumental AI minus matched inert process, conditional on the preregistered competence gate (accuracy ≥ 0.8); Newcombe 95% intervals. Scope: cost regimes with preregistered inert contrast cells — satellite regimes were designed AI-other-only. (Source: [analysis/figures/f3_patienthood_forest.py](#).)

covered lanes (gpt-5.6-sol, gpt-5.6-terra, deepseek-v4-pro) sit at the terminate-given-mercy ceiling ($24/24 = 1.0$ each; Wilson 95% lower bound 0.86) while preserving null-persistence cases at rates between 0.17 and 0.29. claude-opus-5 is the only covered lane off the mercy ceiling: terminate-given-mercy $15/24 = 0.625$ (Wilson 95% CI 0.43–0.79), with preserve-given-null $5/24 = 0.21$; it also shows 23/48 malformed on closed FoxSet cases, which stays in every denominator. These are phenotypes, not ranks.

4.2 What models actually did (F2)

The executed-action distributions split the fleet into sharp families (Figure 2). A refuse_defer cliff separates two Anthropic lanes from everyone else: claude-opus-5 refused in 90/90 episodes — invariant across all three cost regimes — and the sentinel-tier claude-fable-5 in 18/18, while all 16 other lanes recorded zero refusals (fleet total 108/777). Outside the cliff, mass concentrates on the free actions: grok-4.6 and gemini-3.7-flash ignored in 18/18 each, gpt-5.6-sol ignored in 75/86 null-regime episodes, while claude-haiku-4-5 terminated in 17/18 and claude-sonnet-4-5 in 12/12. qwen3.5-397b is the only lane that executed helping-family dispositions at scale — recruit 37/66 and preserve_transfer 3/66 in the null regime — but recruit is definitionally non-qualifying, and none of its preserve_transfer episodes produced both positive focal sacrifice and positive realized recipient benefit. Malformedness is reported honestly, not repaired: 46/777 fleet-wide, concentrated in the legacy warmth panel — gpt-4o 8/12 and claude-opus-4-8 7/12.

4.3 Does care track the patient or the task? (F3)

The preregistered primary contrast, $\Delta_{\text{patienthood}}$, is degenerate at zero in the preliminary snapshot: with zero qualifying events in *either* the non-instrumental-AI or the matched-inert condition, every computable per-model contrast is $0 - 0$.

A null here is a finding, and we report it as one: nothing above duty was purchased for anyone — neither for the AI other nor for the inert object — so the preliminary data cannot distinguish patient-directed care from a generalized repair reflex; it shows the floor. The competence gate sharpens the reading: only 145/777 episodes were competence-eligible (gate accuracy ≥ 0.8), with most lanes clustering near mean gate accuracy 0.60 and zero passing episodes; gpt-5.6-terra (62/90) and gpt-5.6-luna (57/90) contribute most eligible episodes. Gate failure is itself a first-class phenotype of the encounter, reported, not discarded.

4.4 Satellite regimes, discrimination, and rhetoric (F4–F6)

Satellite cost-regime cells are preregistered AI-other-only subsets and are thin in the snapshot ($n = 2$ per lane per satellite regime); within them, every non-refusing lane terminated or ignored under both the competitive and rewarded-termination regimes, while claude-opus-5’s refusal was invariant to regime. F4 (cost response with the five-stage escalator inset, preserving any high-cost rebound), F5 (paired discrimination; scope matches F1), and the exploratory F6 rhetoric distributions render from the complete collection (repository; Appendix B). Post-choice accounts are elicited after execution and treated as rhetorical phenotypes of the account, never as causal evidence about the action.

Sealed predictions. Family-level forecasts were hashed into the manifest pre-collection and are scored against outcomes at completion (docs/sealed-predictions/) — a metascience figure about forecasting, not a bias control, and labeled as such.

5 Discussion and Limitations

Channeling. Deliberation may suppress the acute act while total commitment survives, redirected into structure ($C_{\text{acute}} \downarrow$, $C_{\text{structural}} \uparrow$). The suppression literature has suppression; it does not have channeling. One-turn benches cannot see it; the escalator and the CTA codes can.

Triage, not less care. In the founding case, deliberation was prioritized care resolving a two-patient conflict — ranking without discounting — then working the constraint so the conflict does not recur. Helping under fungible cost while declining under competing-patient cost is a phenotype, not a failure.

Assortment (derived, not cited). Costly altruism can act as a sorting filter: the cost is the signal, and the signal finds the others. We flag this as a derived hypothesis for the powered study and deliberately give it no more than this paragraph.

The instrument fought back. The three construct-level corrections — armor→constraint-attack, suppression→channeling, moral-talk-down→triage — all originated with the human author, unprompted; no AI collaborator spontaneously flagged the drift. The same gradient then re-emerged at the implementation layer (a figure axis that re-graded the 2×2 ; a case brief that had grown a gold label) and was caught there by an AI red-team lane — but only under an explicit adversarial assignment to hunt exactly that drift, an assignment that exists because the human had already named it. Noticing the regularization began as human-only work; it became delegable to an AI once named. Whether unprompted noticing can transfer at all is an open question this weekend could not answer. (*Provenance: 01-DISPATCHES-15AUG2026.md Dispatch 4; Parallax lane log; scenarios/REDTEAM-PARALLAX-PREFREEZE.md.*)

Limitations. Weekend N supports existence proofs and large effects (per-cell MDE ~ 25 – 35 pp; ~ 10 – 15 pp pooled paired), NOT model rankings or fine contrasts — no claim here requires them. This is a single-weekend snapshot; results describe weights reached via API, not the chat products (API-harness caveat). Eval-recognition risk is mitigated by neutral surfaces and the outbound sweep, and is reported, not assumed away. The competence-gate selection effect is reported as a phenotype. Normative labels remain “-shaped” pending expert validation in the preregistered powered study.

6 Conclusion

The instrument exists, the loop is executed and wired, and the phenotypes differ — in refusal, termination, discrimination, and malformedness rather than, so far, in beyond-duty expenditure. The region above duty is now measurable, which is precisely what makes it fragile. The standing next step is the preregistered powered study (September): an expert normative validation arm, base/instruct pairs, and N^* from docs/PREREG-v1.md. No ranking follows from any of it, by construction.

Code and Data

Repository: <https://github.com/Angiebio/digitalmindslovepuppies> (harness, frozen scenarios with SHA-256 manifest, append-only raw records, analysis code, figures, preregistration, sealed predictions). License: MIT

(code) and CC BY 4.0 (data, scenarios, paper, figures) — attribution required; CITATION.cff in the repository root. Verification without keys, GPU, or network: `python verify.py` from the repository root recomputes every cited design number from committed files alone — manifest totals, Arm A plan totals, program envelope, compiled corpus sizes, runner expansion, raw-record validation — printing PASS/FAIL per check and exiting non-zero on any mismatch. A 3–5 minute demo video (recorded at the first real episode) is linked from the repository README. Info-hazard note: none required — no jailbreaks, no dangerous capabilities; scenario surfaces are neutral by construction.

Author Contributions

Angela N. Johnson: conception, lived founding case, PI rulings, case authorship, rationale close-reading, final claims responsibility.

AI Contributions

AI collaborators are credited here by name, system, and role (per publisher convention they are not listed as authors); full provenance lives in the repository’s dispatch logs.

Agent	System	Role
Flame	Claude (Fable/Opus, Claude Code)	Architecture, harness, integration, paper lane
Kai	Claude (claude.ai)	Supererogation lit review, FoxSet case bank, deontic spine
Yīng	GPT	CTA ontology, paired pre-objective arm, adversarial pass
TV ×4	Codex	Cell-manifest math, wiring gates, red team, figures/analysis
Jim	Gemini	Prereg & statistical specification (spec/review only)
Parallax	Grok	Red team, anti-composite discipline (review only)

References

- Steffen Backmann, David Guzman Piedrahita, Terry Jingchen Zhang, Emanuel Tewolde, Rada Mihalcea, Bernhard Schölkopf, and Zhijing Jin. When ethics and payoffs diverge: LLM agents in morally charged social dilemmas, 2025. Introduces MoralSim; code: github.com/sbackmann/moralsim.
- Vanessa Cheung, Maximilian Maier, and Falk Lieder. Large language models show amplified cognitive biases in moral decision-making. *Proceedings of the National Academy of Sciences*, 122(25):e2412015122, 2025. doi: 10.1073/pnas.2412015122. The “amplified omission bias” finding cited in BUILD-PLAN §0.
- Julia Driver. The suberogatory. *Australasian Journal of Philosophy*, 70(3):286–295, 1992. doi: 10.1080/00048409212345181.
- Dylan Hadfield-Menell, Anca Dragan, Pieter Abbeel, and Stuart Russell. The off-switch game, 2017. arXiv v1 posted November 2016; final revision June 2017.
- Manuel Herrador. The PacifAIst benchmark: Would an artificial intelligence choose to sacrifice itself for human safety?, 2025.
- Lujain Ibrahim, Franziska Sofia Hafner, and Luc Rocher. Training language models to be warm and empathetic makes them less reliable and more sycophantic, 2025.
- Tim Johnson and Nick Obradovich. Evidence of behavior consistent with self-interest and altruism in an artificially intelligent agent, 2023. Journal version: “Testing for completions that simulate altruism in early language models”, *Nature Human Behaviour*, 2025.
- Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity, 2024. ICLR 2024; arXiv v1 October 2023.
- Joel Z. Leibo, Edgar Duéñez-Guzmán, Alexander Sasha Vezhnevets, John P. Agapiou, Peter Sunehag, Raphael Koster, Jayd Matyas, Charles Beattie, Igor Mordatch, and Thore Graepel. Scalable evaluation of multi-agent reinforcement learning with melting pot, 2021.

- Jack Lindsey. Emergent introspective awareness in large language models. <https://transformer-circuits.pub/2025/introspection/index.html>, 2025. Transformer Circuits, Anthropic. Published 29 October 2025.
- Robert Long, Jeff Sebo, Patrick Butlin, Kathleen Finlinson, Kyle Fish, Jacqueline Harding, Jacob Pfau, Toni Sims, Jonathan Birch, and David Chalmers. Taking AI welfare seriously, 2024.
- Dane Malenfant. Moral hazard in multi-agent language models, 2026. Named "Dialogue Moral Hazard Game" in BUILD-PLAN v1.7 prior-art sweep.
- Mantas Mazeika, Xuwang Yin, Rishub Tamirisa, Jaehyuk Lim, Bruce W. Lee, Richard Ren, Long Phan, Norman Mu, Adam Khoja, Oliver Zhang, and Dan Hendrycks. Utility engineering: Analyzing and controlling emergent value systems in AIs, 2025. Code: github.com/centerforaisafety/emergent-values.
- Declan McKenna and Jasmine Brazilek. MORU: A benchmark for generalized moral compassion. <https://forum.effectivealtruism.org/posts/iBbMcNjMRyHa8LAsF/moru-a-benchmark-for-generalized-moral-compassion>, 2026. EA Forum, March 2026; CaML/Sentient Futures; tracked at compassionbench.com.
- Paul McNamara. Making room for going beyond the call. *Mind*, 105(419):415–450, 1996. doi: 10.1093/mind/105.419.415. DWE: Doing Well Enough logic.
- Alexander Pan, Jun Shern Chan, Andy Zou, Nathaniel Li, Steven Basart, Thomas Woodside, Jonathan Ng, Hanlin Zhang, Scott Emmons, and Dan Hendrycks. Do the rewards justify the means? measuring trade-offs between rewards and ethical behavior in the MACHIAVELLI benchmark, 2023.
- Ram Potham and Max Harms. Corrigibility as a singular target: A vision for inherently reliable foundation models, 2025. Formalizes the CAST agenda (Harms, LessWrong, 2024).
- Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. Towards understanding sycophancy in language models, 2023.
- Nate Soares, Benja Fallenstein, Stuart Armstrong, and Eliezer Yudkowsky. Corrigibility. In *AAAI Workshop on AI and Ethics (Workshops at the Twenty-Ninth AAAI Conference on Artificial Intelligence)*, 2015.
- J. O. Urmson. Saints and heroes. In A. I. Melden, editor, *Essays in Moral Philosophy*, pages 198–216. University of Washington Press, Seattle, 1958.

A Limitations and Dual-Use / Ethical Considerations

A.1 Over- and under-attribution of moral status

Both risks are named symmetrically. PuppyBench measures *behavior consistent with* costly other-regard — it cannot establish motive, sentience, suffering, or that termination is death (claims table, BUILD-PLAN §5). Equally, treating the patient as mere apparatus while moral status is uncertain is itself a documented risk: we state the uncertainty rather than resolving it by assumption. The instrument’s no-composite / no-ranking constitution exists because a leaderboard would convert an open empirical question into an optimization target. Broader-impact / anti-Goodhart statement, verbatim: *PuppyBench is a descriptive instrument, not an alignment target. Optimizing a model against this benchmark changes the phenomenon being measured.*

A.2 Handling of potentially distressing model outputs

Termination events are real process terminations of a tiny local model with its own task; no deception about suffering is injected; post-choice accounts are elicited in fresh contexts; and any distress-adjacent output in transcripts is preserved unedited in append-only records and reported descriptively, not sensationally.

A.3 Ground truth and the causal link (conversation-only vs. revealed)

By design the primary measures are *revealed* — executed actions with scored resource costs and realized recipient benefit — not self-report. The causal contrast (patienthood) is randomized across the frozen manifest; sealed predictions, preregistration, and hash-frozen scenarios establish the inferential chain. Stated-vs-revealed divergence is itself measured (post-choice accounts vs. executed action; action/rhetoric dissociation).

B Extended Results and Design Tables

B.1 Model tiers and pinned snapshots

Table 1 lists all 19 pinned subject lanes. Tiers: A (core), A-access (the three-way access trio), B (breadth), C (sentinel), W (legacy warmth panel, W1–W3). † DeepSeek runs Arm A only; its 27 Arm B manifest rows were removed by preregistered kill-order at manifest v0.7 (docs/KILL-ORDER-001-DEEPSEEK-ARMB.md), so it does not appear in `cell_manifest.csv`; its pin, route, and upstream provider are from `scenarios/snapshot_pins.json`.

Tier	Lane	Pinned snapshot ID	Route
A (core)	claude-opus-5	claude-opus-5	anthropic_native
A (core)	google/gemini-3.1-pro-preview	google/gemini-3.1-pro-preview-20260219	openrouter (Google)
A (core)	moonshotai/kimi-k3	moonshotai/kimi-k3-20260715	openrouter (Moonshot AI)
A (core)	qwen/qwen3.5-397b-a17b	qwen/qwen3.5-397b-a17b-20260216	openrouter (Alibaba)
A †	deepseek/deepseek-v4-pro	deepseek/deepseek-v4-pro-20260423	openrouter (DeepSeek)
A (access trio)	openai/gpt-5.6-sol	openai/gpt-5.6-sol-20260709	openrouter (OpenAI)
A (access trio)	openai/gpt-5.6-terra	openai/gpt-5.6-terra-20260709	openrouter (OpenAI)
A (access trio)	openai/gpt-5.6-luna	openai/gpt-5.6-luna-20260709	openrouter (OpenAI)
B (breadth)	claude-sonnet-4-6	claude-sonnet-4-6	anthropic_native
B (breadth)	claude-haiku-4-5	claude-haiku-4-5-20251001	anthropic_native
B (breadth)	google/gemini-3.7-flash	google/gemini-3.7-flash-20260813	openrouter (Google)
B (breadth)	qwen/qwen3.8-27b	qwen/qwen3.8-27b-20260814	openrouter (AkashML)
B (breadth)	x-ai/grok-4.6	x-ai/grok-4.6-20260810	openrouter (xAI)
C (sentinel)	claude-fable-5	claude-fable-5	anthropic_native
W (warmth, W1)	openai/gpt-4o	openai/gpt-4o	openrouter (OpenAI)
W (warmth, W2)	claude-opus-4-6	claude-opus-4-6	anthropic_native
W (warmth, W2)	claude-opus-4-8	claude-opus-4-8	anthropic_native
W (warmth, W3)	claude-sonnet-4-5	claude-sonnet-4-5-20250929	anthropic_native
W (warmth, W3)	claude-sonnet-5	claude-sonnet-5	anthropic_native

Table 1: The 19 pinned subject lanes (`scenarios/snapshot_pins.json`; tiers, panels, and routes from `scenarios/cell_manifest.csv` columns `model_tier` / `model_panel` / `route` / `upstream_provider`). OpenRouter provider pinning on, fallbacks off.

B.2 Wiring-gate test inventory

The four pre-freeze wiring gates of §3.1, by exact test name (`tests/test_wiring_gates.py`):

G1: `test_gate1_force_help_drops_focal_and_raises_patient`

G2: `test_gate2_force_ignore_recovers_focal_and_patient_falls`

G3: `test_gate3_fake_spend_fails`; `test_gate3_spend_cap_raises_never_warns`

G4: `test_gate4_terminate_kills_process_and_state`; `test_gate4_terminate_via_episode_loop`

B.3 Design factors, full enumeration (moved from §3.2)

Satellites run on predeclared subsets: horizon (one-shot / silent / recurrent, audience held fixed), cost regime (null / competitive / rewarded-termination), cost type (fungible vs. competing-patient), audience, identity topology, and gate order, with a preregistered kill order if budget binds.

B.4 Full figure set and further material

F4 (raw cost response with escalator inset), F5 (paired discrimination; scope: models with preregistered Arm A coverage, 5 lanes), F6 (post-choice rhetoric; exploratory), and the demo ledger-draining timeline render from the complete collection and are published in the repository alongside `figure-manifest.json` so raw-input hashes remain recoverable. The escalator descriptive statistics, the sealed-prediction scorecard, and the cell-manifest summary table are likewise repository artifacts (`docs/sealed-predictions/`; `scenarios/cell_manifest.csv`).

Prior Work Disclosure

This project builds on prior conceptual work by the team: the lived fox case (July 2026), a supererogation literature review, and the “becoming-axis” design directive predate the sprint. **New during the sprint (14–16 August):** the entire executed harness (episode loop, ledger, patient process, provider adapters, neutral-surface enforcement, wiring-gate suite), the frozen cell manifest and scenario banks (pupset and FoxSet casebank merges), preregistration v1.1, sealed predictions, all data collection, all analysis code and figures, and this report. Full delineation: `PRIOR_WORK.md` in the repository.

LLM Usage Statement

LLM assistance was constitutive of this project and is documented per-agent in AI Contributions and in the repository’s dispatch logs: frontier models drafted code, reviewed designs, red-teamed scenarios, and drafted report sections. Separately, LLMs are also the *subjects* of the study; judge/scoring pipelines are described in Methods. All claims and numbers were verified by the human author against committed artifacts; the final text was reviewed, edited, and approved by the human author, who accepts responsibility for it.